ORIGINAL ARTICLE

doi:10.1111/evo.12505

The influence of life-history strategy on genetic differentiation and lineage divergence in darters (Percidae: Etheostomatinae)

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Received January 27, 2014

Accepted July 16, 2014

Recent studies determined that darters with specialized breeding strategies can exhibit deep lineage divergence over fine geographic scales without apparent physical barriers to gene flow. However, the extent to which intrinsic characteristics interact with extrinsic factors to influence population divergence and lineage diversification in darters is not well understood. This study employed comparative phylogeographic and population genetic methods to investigate the influence of life history on gene flow, dispersal ability, and lineage divergence in two sympatric sister darters with differing breeding strategies. Our results revealed highly disparate phylogeographic histories, patterns of genetic structure, and dispersal abilities between the two species suggesting that life history may contribute to lineage diversification in darters, especially by limiting dispersal among large river courses. Both species also showed striking differences in demographic history, indicating that extrinsic factors differentially affected each species during the Pleistocene. Collectively, our results indicate that intrinsic and extrinsic factors have influenced levels of gene flow among populations within both species examined. However, we suggest that life-history strategy may play a more important role in lineage diversification in darters than previously appreciated, a finding that has potentially important implications for understanding diversification of the rich North American freshwater fish fauna.

KEY WORDS: **Dispersal ability, genetic structure, microsatellites, phylogeography, sister species.**

Understanding mechanisms that influence genetic structure of populations within species can provide valuable insights into lineage diversification, and ultimately speciation (Bohonak 1999; Coyne and Orr 2004). Patterns of lineage diversification in freshwater taxa are often unique from other organisms because they are confined to aquatic habitats of the drainage basins in which they live (Avise 2000; Bilton et al. 2001; Berendzen et al. 2008). Thus, extrinsic factors such as interdrainage divisions and connections, underlying geologic features, habitat features, and the distribution of other organisms have been deservedly paramount in the explanation of lineage diversification in freshwater taxa (Wiley and Mayden 1985; Starnes and Etnier 1986; Mayden 1988; Near et al. 2001; Marten et al. 2006; Burridge et al. 2008). Additionally, intrinsic characteristics of freshwater taxa such as reproductive strategy, fecundity, dispersal ability, and body size are thought to influence levels of gene flow and thus geographic patterning of diversification (Tibbets and Dowling 1996; Turner et al. 1996; Turner and Trexler 1998; Whiteley et al. 2004; Cook et al. 2006; Steele et al. 2009). Despite numerous studies on the influence of life-history characteristics on geographic patterns of evolution (Yu et al. 2010; Dawson 2012), the associations between intrinsic factors, extrinsic factors, and lineage divergence in freshwater fishes remain poorly understood.

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The freshwater fish fauna of eastern North America is one of the most diverse assemblages of any temperate region worldwide (Lundberg et al. 2000; Abell et al. 2008). Of this fauna, perhaps no other clade of fishes has been more thoroughly studied in terms of the effects of extrinsic barriers on diversification than darters (Teleostei: Percidae: Etheostomatinae). Darters are a specious group of small, benthic fishes restricted to freshwaters of eastern North America (Page 1983). With approximately 250 species, darters represent one of the most successful evolutionary radiations in North America making them an exceptional model system in the study of freshwater fish diversification (Page 1983; Boschung and Mayden 2004; Keck and Near 2009; Near et al. 2011; Kelly et al. 2012). The high diversity of darters and other eastern North American fishes was largely shaped by allopatric isolation in disjunct geographic areas and neighboring river systems (Wiley and Mayden 1985; Starnes and Etnier 1986; Mayden 1988; Berendzen et al. 2008; Hollingsworth and Near 2009; Keck and Near 2010). Similarly, population genetic differentiation in many darter species is partitioned according to extrinsic features such as physiography and drainage basin (Piller et al. 2008; Powers and Warren 2009; Fluker et al. 2010; Austin et al. 2011). However, recent studies within the geologically stable Cumberland River Drainage revealed that darters can exhibit remarkable levels of diversification over extremely fine geographic scales (microendemism) with no obvious extrinsic barriers to gene flow. For example, Hollingsworth and Near (2009) recovered five deeply divergent clades (time to most common recent ancestor [TMRCA] 2−8 million years ago [mya]) within *Etheostoma basilare*, a species complex endemic to tributaries within a single river drainage. Keck and Near (2010) found a similar, but more recent and widespread pattern of microendemism in *Nothonotus* darters of the Cumberland River, which they attributed to niche conservatism. These newly highlighted patterns of within-drainage diversification in darters, despite obvious extrinsic barriers to gene flow among populations, indicate that differentiation must be driven by currently unknown physical or nonphysical barriers to gene flow.

Several studies have determined that life-history characteristics exert a significant influence on gene flow, and thus lineage diversification, in freshwater fishes (Tibbets and Dowling 1996; Turner et al. 1996; Turner and Trexler 1998; Whiteley et al. 2004). Although these studies were elegant in the use of sympatric, closely related species in analysis of intrinsic factors on lineage divergence, they did not include sympatric sister species comparisons. Undoubtedly, finding sympatric sister species with clear differences in life-history attributes is particularly difficult regardless of taxonomic group (Steele et al. 2009; Dawson 2012). This is especially true for North American freshwater fishes, in which sister species are often distributed allopatrically among geographic regions or drainage basins (Hollingsworth and Near 2009), and where sisters are sympatric they rarely exhibit differing life-history traits (Page 1985).

Two darter species that meet the optimal criteria of being sympatric sisters with strikingly different life-history strategies are members of the darter subgenus *Psychromaster* (sensu Near et al. 2011): the Slackwater Darter (*Etheostoma boschungi*) and the Tuscumbia Darter (*E. tuscumbia*). Although these two species are not syntopic, they share an overlapping distribution in tributaries to the lower bend of the Tennessee River (Figs. 1, 2). Their sister relationship was only recently uncovered by molecular phylogenetic studies (Mayden et al. 2006; Lang and Mayden 2007; Near et al. 2011), and was previously obscured by their prominent differences in morphological and life-history characteristics. Excluding *E. tuscumbia*, all members of *Psychromaster* (*E. autumnale, E. boschungi*, *E. cragini*, *E. mihileze, E. pallididorsum*, and *E. punctulatum*) and *E. trisella* were previously considered members of the subgenus *Ozarka* (Williams and Robison 1980), a group of darters defined by unique male breeding coloration and reproductive strategy. Of particular interest is the breeding strategy of *E. boschungi* and other members of "*Ozarka*," which is unique to darters. During nonspawning times, *E. boschungi* inhabits gentle riffles and slackwater areas of small to medium sized upland streams. However, for relatively brief periods during winter and early spring, they migrate into tiny spring-fed rivulets or ephemeral seeps, spawn, and subsequently return to the stream (Hambrick and Robison 1979; Williams and Robison 1980; Boschung and Nieland 1986; Bailey and Etnier 1988). In contrast to *E. boschungi* and other "*Ozarka*," *E. tuscumbia* lacks male breeding coloration and does not exhibit the brief annual breeding migrations into ephemeral spring-fed habitats. Rather, *E. tuscumbia* inhabits isolated limestone spring pools throughout its range where it is rarely found in streams or beyond persistent groundwater influence (Armstrong and Williams 1971; Page 1983; Boschung and Mayden 2004). Because of its thermally stable spring habitats, some populations of *E. tuscumbia* have protracted spawning periods, or spawn continuously throughout the year (Koch 1978; Boyce 1997).

In this study, we employ comparative phylogeographic and population genetic methods to examine the effect of differing life-history strategies on phylogeographic structure and gene flow among populations of *E. boschungi* and *E. tuscumbia*. In comparative phylogeography, assessing genealogical concordance among co-distributed lineages provides a powerful tool for elucidating mechanisms or barriers to dispersal driving the process of diversification (Avise 2000; Zink 2002). If phylogeographic histories are concordant among co-distributed species, they have likely responded in parallel to shared environmental or geologic conditions and have common extrinsic barriers to gene flow (Riddle et al. 2000; Irwin 2002). However, if phylogeographic

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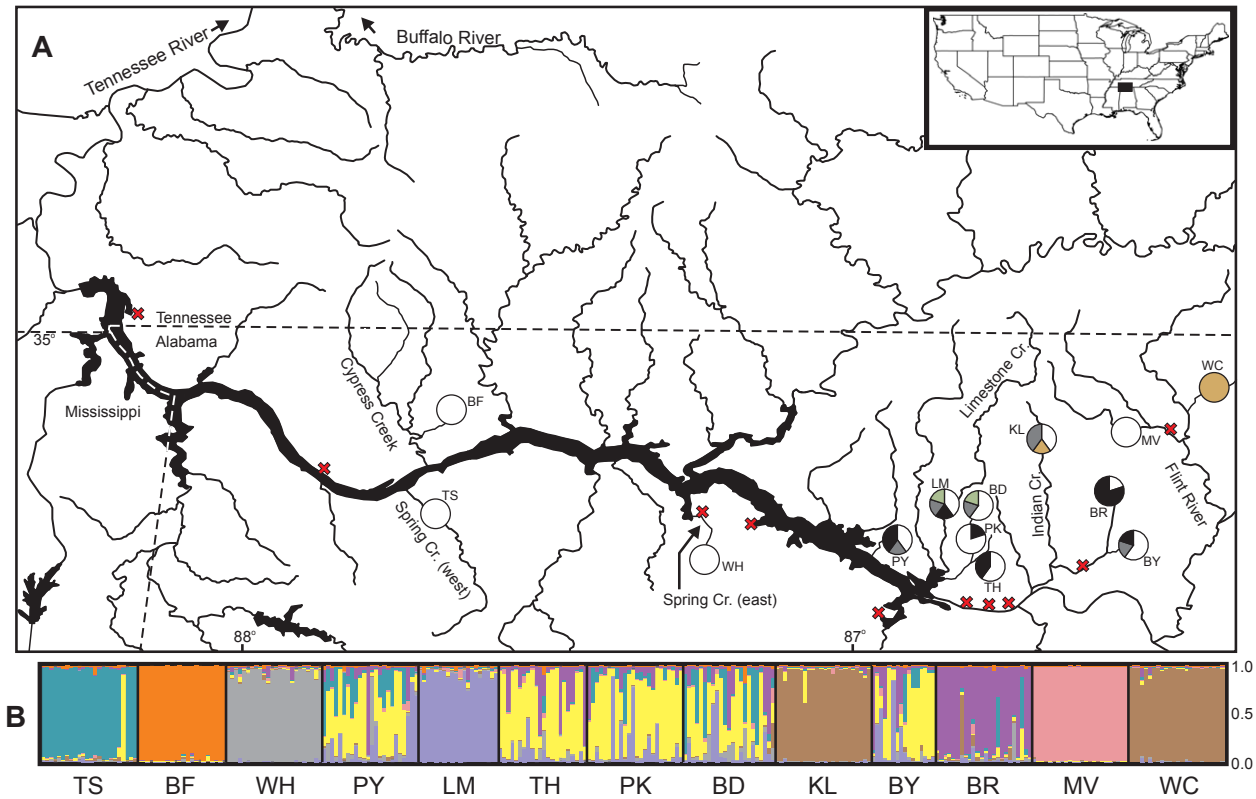

Figure 1. (A) Map of the lower bend of the Tennessee River showing sample locations for *Etheostoma tuscumbia*. Locality codes correspond to Table 1. Pie charts represent mitochondrial DNA haplotype endemicity, where white represents the proportion of haplotypes unique to the locality and colors represent the proportion shared with other localities. Red X indicates extirpated localities. (B) Results from the microsatellite DNA-based STRUCTURE analysis of *E. tuscumbia*, showing the most likely genetic structure ($K = 8$). Bars correspond to multilocus genotypes of individuals and colors represent the probability of ancestry to each cluster (K).

histories are discordant, insights into intrinsic, lineage-specific characteristics such as dispersal ability can be gained (Hodges et al. 2007; Papadopoulou et al. 2009).

Given the shared distribution of *E. boschungi* and *E. tuscumbia* in an area of long-term geologic stability that has been unaffected by Pleistocene glacial ice sheets, and both species' strict reliance on groundwater sources, we test the null hypothesis that both species will have concordant phylogeographic structures. Failure to reject this hypothesis would indicate that life-history strategy has little or no influence on phylogeographic structuring within these two species, and that extrinsic features play a key role in shaping patterns of diversification. Alternatively, incongruent phylogeographic histories may provide evidence that intrinsic characteristics are important in shaping patterns of diversification in these two species.

Another objective of this study was to evaluate the effect of life-history differences on gene flow between populations at differing spatial scales within a single river system. For example, large river courses appear to be barriers to dispersal for species with strict breeding habitat requirements or breeding habits in small tributaries, such as the *E. basilare* complex (Hollingsworth and Near 2009) and *E. pallididorsum* (Turner and Robison 2006). We used mitochondrial (mt) DNA sequence data and microsatellite DNA loci to estimate genetic structure, gene flow, and demographic history for both *E. boschungi* and *E. tuscumbia* throughout their range. These data were used to evaluate how differences in life-history strategy influence gene flow among populations within a single tributary to the major river course (i.e., the Tennessee River) versus among populations in distinct tributaries of the major river.

Materials and Methods

SAMPLE COLLECTION AND DNA EXTRACTION

We obtained tissue samples from across the entire range of *E. boschungi* ($N = 149$) and *E. tuscumbia* ($N = 295$) in tributaries to the lower bend of the Tennessee River (Figs. 1, 2; Table 1). Samples were collected by seine, or obtained from the University of Alabama Ichthyology Collection (UAIC) frozen

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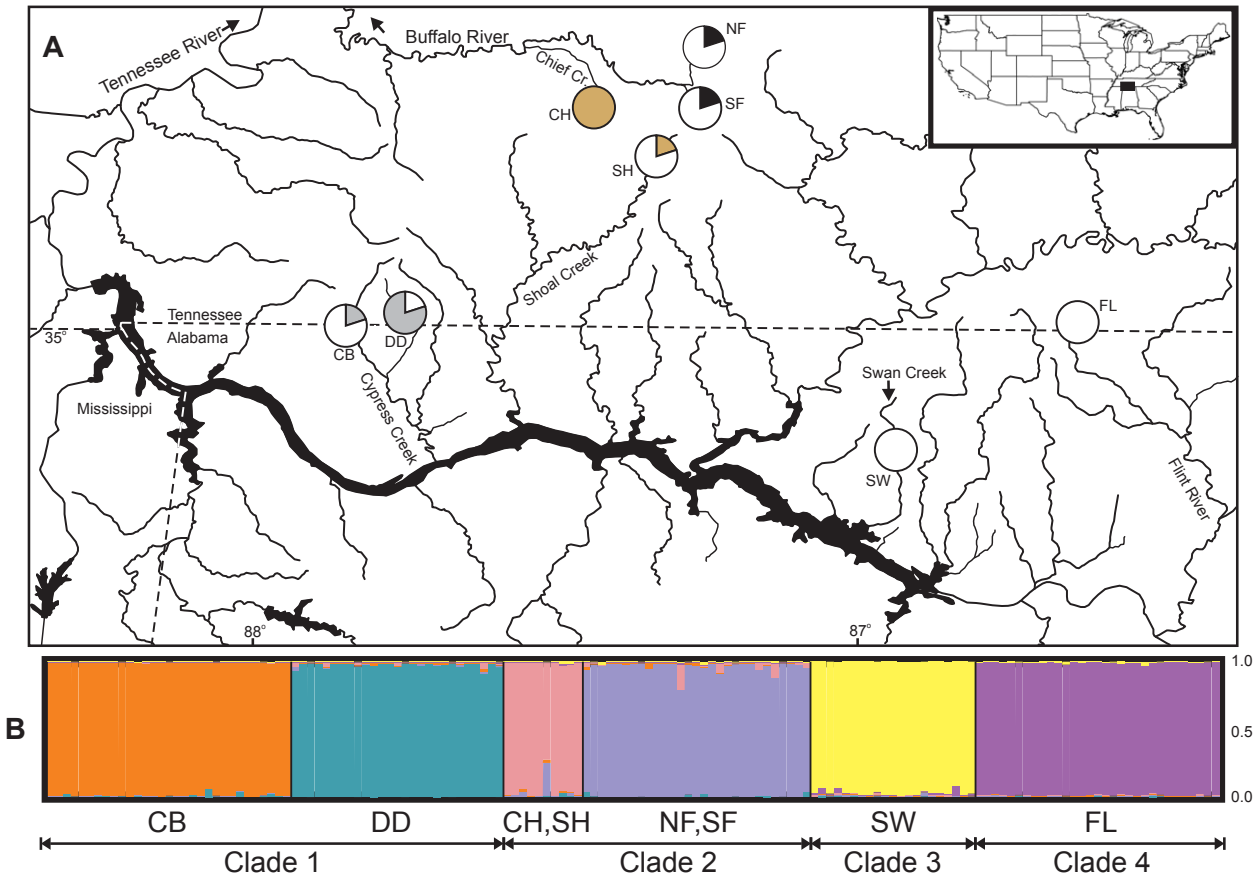

Figure 2. (A) Map of the lower bend of the Tennessee River showing sample locations for *Etheostoma boschungi*. Locality codes correspond to Table 1. Pie charts represent mitochondrial DNA haplotype endemicity, where white represents the proportion of haplotypes unique to the locality and colors represent the proportion shared with other localities. (B) Results from the microsatellite DNA-based STRUCTURE analysis of *E. boschungi*, showing the most likely genetic structure ($K = 6$). Bars correspond to multilocus genotypes of individuals and colors represent the probability of ancestry to each cluster (K). Clades correspond to Figure 3.

tissue collection (Table S1). Due to the difficulty of collecting *E. boschungi* outside of breeding sites and putative decrease in abundance at several extant breeding sites (Boschung and Nieland 1986; Rakes and Shute 2008; Johnston et al. 2013), two of the eight known extant breeding sites yielded only five individuals each despite multiple attempts over a four-year period. Individuals of *E. tuscumbia* were obtained from 12 of the 14 extant spring localities and a newly discovered site in Limestone Creek. Tissues and voucher specimens taken in the field were preserved in 95% ethanol and 10% formalin, respectively, and deposited into UAIC (Table S1). Genomic DNA was extracted from tissues using the DNeasy kit (Qiagen Inc., Valencia, CA).

DNA SEQUENCING AND MICROSATELLITE GENOTYPING

Material for phylogenetic analyses was amplified for the complete mtDNA NADH dehydrogenase subunit 2 (ND2) gene (Table 1). A subset of individuals was amplified for the first intron of the nuclear ribosomal protein S7 (Table 1). Polymerase chain reaction (PCR) amplifications for both loci followed protocols of Lang and Mayden (2007). The resulting products were sequenced at the Steven Johnson Molecular Systematics Laboratory, University of Alabama. DNA sequences were deposited in GenBank (Table S1).

All individuals were genotyped for nine microsatellite loci using primers designed for other darters (Eca10EPA, Eca46EPA, Eca48EPA [Tonnis 2006]; EosC3, EosC6, EosD11, EosD107 [Switzer et al. 2008]; Esc26b, Esc120 [Gabel et al. 2008]). PCR amplifications followed protocols of Fluker et al. (2010). Fluorescently labeled fragments were read on an ABI 3730 DNA analyzer (University of Maine DNA sequencing facility). Prior to analyses, alleles were binned using the program FLEXIBIN 2.0 (Amos et al. 2007) and the dataset was screened for genotyping errors with MICRO-CHECKER (van Oosterhout et al. 2004).

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Table 1. Details of *Etheostoma boschungi* and *E. tuscumbia* specimens used in this study showing locality ID (see Figs. 1 and 2), number of individuals genotyped for microsatellite loci (*N* microsat), number of individuals sequenced for the mitochondrial ND2 gene (*N* mtDNA) and nuclear intron S7 (*N* S7), and mtDNA haplotype code.

Species/drainage/locality	ID	*N* microsat	*N* mtDNA	*N* S7	Haplotype
Etheostoma boschungi					
Cypress Creek					
Cooper Branch	CB	31	5	–	1, 2, 3, 4
Dodd Site	DD	27	5	1	1, 5
Shoal Creek	SH	5	5	1	6, 7
Buffalo River					
Chief Creek	CH	5	5	–	6
North Fork	NF	18	5	–	8, 9
South Fork	SF	11	5	–	8, 10, 11
Swan Creek	SW	21	5	1	12, 13, 14
Flint River					
Brier Fork	FL	31	6	1	15, 16, 17
E. tuscumbia					
Spring Creek (west)					
Tuscumbia Spring	TS	24	5	–	18, 19, 20, 21
Cypress Creek					
Buffler Spring	BF	22	5	1	22, 23
Spring Creek (east)					
Wheeler Spring	WH	24	5	–	24
Pryor Branch					
Pryor Spring	PY	24	5	–	25, 26, 27, 28
Limestone Creek					
Limestone Creek	LM	20	5	–	25, 26, 29, 30, 31
Thorsen Spring	TH	22	5	–	25, 32
Pickens Spring	PK	24	5	–	25, 33, 34
Beaverdam Spring	BD	23	5	–	26, 29, 35, 36
Indian Creek					
Kelly Spring	KL	24	5	–	26, 37, 38
Byrd Spring	BY	16	5	–	25, 26, 39, 40
Braham Spring	BR	24	5	–	25, 41
Flint River					
Meridianville Spring	MV	24	5	1	42
Unnamed Spring	WC	24	5	–	37

Final microsatellite data files available from the DRYAD digital repository (http://doi.org/10.5061/dryad.2s07g).

PHYLOGEOGRAPHIC RECONSTRUCTION

Unique mtDNA haplotypes were determined for both species using the program DnaSP version 5.10 (Librado and Rozas 2009) and were aligned with darter outgroups obtained from GenBank, including members of "*Ozarka*" (Table S1). Best-fit models of nucleotide substitution were evaluated for individual codon positions of ND2 using Akaike information criterion in the program MrModeltest version 2.3 (Nylander 2004). To reconstruct phylogeographic histories of both species, Bayesian phylogenetic inference was performed using MrBayes version 3.12 (Ronquist and Huelsenbeck 2003) on the ND2 dataset. Codon positions were partitioned and the appropriate substitution models were used to set unlinked priors for each (Table S2). Four separate runs were conducted for 20 million generations with sampling every 1000 generations. Convergence of runs was assessed with the program Tracer version 1.5 and by monitoring the standard deviation of the split frequencies. All runs prior to convergence (>0.01) were discarded as burnin to construct the 50% majority-rule consensus phylogram. To visualize genealogical relationships among mtDNA sequences, haplotype networks were constructed using statistical parsimony in TCS version 1.21 (Templeton et al. 1992; Clement et al. 2000). TCS constructs haplotype relationships by calculating the maximum number of substitutions to connect two haplotypes parsimoniously with 95% confidence. TCS also estimates outgroup probabilities, identifying probable roots and

the most ancient haplotypes in the network (Castelloe and Templeton 1994).

DIVERGENCE TIME ESTIMATES

The second alignment consisted of concatenated ND2-S7 sequences to estimate divergence times among clades of *E. boschungi* and *E. tuscumbia*. Because good darter fossils are lacking (Cavender 1986), we employed external fossil calibration methods of Hollingsworth and Near (2009), which utilized DNA sequence data and five fossil calibration points for the closely related family Centrarchidae (Near et al. 2005). Preliminary phylogeographic analysis recovered four highly divergent, well-supported clades within *E. boschungi* and shallow divergence within *E. tuscumbia*. Thus, the final alignment consisted of one concatenated sequence from each of the four *E. boschungi* clades, one *E. tuscumbia*, 47 Centrarchid taxa (Near et al. 2005), and the same darter outgroups as the phylogeographic analysis (Table S1). The S7 dataset was aligned using the program MUSCLE version 3.8 (Edgar 2004) and ND2 sequences were aligned by eye.

We estimated divergence times using the program BEAST version 1.8.0 (Drummond and Rambaut 2007) with priors for minimum bound lognormal age estimates on Centrarchid fossil calibration points used in Hollingsworth and Near (2009). Best-fit substitution models for ND2 codon positions and the S7 partition (Table S2) were determined using jModelTest version 2.1.5 (Darriba et al. 2012). To allow for uncertainty in substitutions rates on adjacent branches, an uncorrelated lognormal clock model was employed with a Yule process speciation tree prior. Four independent runs of 60 million generations were conducted and tree and log files were combined with 40% of the generations discarded as burnin using LogCombiner (version 1.8.0, distributed as a part of the BEAST package). Tracer was used to evaluate convergence and stationarity of runs and adequate effective sample size of estimated parameters.

BEAST was also used to estimate the root age for all individuals of *E. tuscumbia*. Best-fit substitution models for ND2 codon positions (Table S2) were determined in MrModeltest version 2.3 (Nylander 2004) from all 65 sequences from *E. tuscumbia*. Four independent runs of 20 million generations were conducted with a coalescent Bayesian Skyline tree prior (Drummond et al. 2005), seven piece-wise intervals, and an estimated rate of ND2 molecular evolution (9.29×10^{-3} substitutions/site/million years). The rate of molecular evolution for ND2 was estimated in a separate BEAST analysis using only ND2 sequences of the 47 Centrarchid taxa and fossil calibration points of Near et al. (2005).

ESTIMATES OF GENETIC DIVERSITY, POPULATION STRUCTURE, AND GENE FLOW

Departures from Hardy–Weinberg equilibrium (HWE) and linkage disequilibrium were assessed for microsatellite loci with GENEPOP version 4.0.10 (Rousset 2008). Measures of genetic diversity were estimated from microsatellites as mean number of alleles per locus (A), heterozygosity observed (H_o), and heterozygosity expected (H_e) using GENEPOP. To provide an estimate of the number of alleles per locus assuming equal sample sizes, allelic richness (AR) was estimated using the program HP-RARE (Kalinowski 2005).

Genetic differentiation was assessed using microsatellites (F_{ST}) and mtDNA (Φ_{ST}) for differing hierarchical groupings using analysis of molecular variance (AMOVA; Excoffier et al. 1992) in ARLEQUIN version 3.5 (Excoffier and Lischer 2010) with 10,000 permutations. To evaluate whether both species exhibit genetic differentiation across their range, we conducted separate AMOVAs for each species, testing the null hypothesis that individuals from each locality represent an arbitrary subsample relative to its entire range. To evaluate whether the Tennessee River has been a significant barrier to dispersal, we conducted separate AMOVAs for each species, testing whether genetic variation was more proportionally distributed among versus within tributaries to the Tennessee River. In the latter test, localities were grouped within tributaries that share a single connection to the Tennessee River. For *E. boschungi*, groups were as follows: (1) Cypress Creek (CB, DD); (2) Shoal Creek (SH, CH, NF, SF); (3) Swan Creek (SW); and (4) Flint River (FL; Fig. 2; Table 1). Samples from the Buffalo River (CH, NF, SF) were grouped with Shoal Creek (SH) because preliminary analyses indicated that Buffalo River populations were recently founded from Shoal Creek. For *E. tuscumbia*, groups were as follows: (1) Spring Creek west (TS); (2) Cypress Creek (BF); (3) Spring Creek east (WH); (4) Pryor Branch (PY); (5) Limestone Creek (LM, TH, PK, BD); (6) Indian Creek (KL, BR, BY); and (7) Flint River (MV, WC; Fig. 1; Table 1).

The program STRUCTURE 2.3 (Pritchard et al. 2000) was used with microsatellite data to determine the number of genetically differentiated clusters (K) within both species without a priori designations. For both species, we implemented models allowing admixture of genotypes and correlated allele frequencies (Falush et al. 2003). Ten replicate STRUCTURE runs of 1,000,000 iterations with 300,000 burnin were conducted for K values ranging from 2 to 10 and 2 to 15 for *E. boschungi* and *E. tuscumbia*, respectively. Best estimates of K for each species were determined using the ad hoc summary statistic ΔK (Evanno et al. 2005) as implemented in STRUCTURE Harvester (Earl 2011).

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We performed coalescent analyses to estimate the parameter Θ (mtDNA $= N_e\mu$, microsatellites $= 4N_e\mu$) and migration rates (M) for mtDNA sequences and microsatellite loci using the Bayesian method in LAMARC version 2.1.6 (Kuhner 2006). We were specifically interested in comparing levels of gene flow among and within tributaries to the Tennessee River for both species. To avoid prohibitively large genealogies and number of parameters, which can lead to poor estimates, we constructed reduced datasets for LAMARC analyses. Estimates of among tributary migration for *E. boschungi* were conducted using the following groups: (1) Cypress Creek (CB); (2) Shoal Creek (SH/CH); (3) Swan Creek (SW); and (4) Flint River (FL; Fig. 2; Table 1). Two independent tests were conducted to evaluate within tributary migration for *E. boschungi*. The first compared localities within Cypress Creek (CB and DD) and the second compared localities within the Buffalo River (NF and SF; Fig. 2; Table 1). We used the following groupings to estimate among tributary migration for *E. tuscumbia*: (1) Cypress Creek (BF); (2) Spring Creek west (WH); (3) Limestone Creek (BD); and (4) Flint River (MV; Fig. 1; Table 1). Three independent analyses were conducted to estimate within tributary migration for *E. tuscumbia*: (1) localities within the Limestone Creek drainage (LM and BD); (2) localities within Indian Creek (KL, BR, and BY); and (3) localities within Flint River (MV and WC; Fig. 1; Table 1). All mtDNA sequences for each locality were used in estimations, while microsatellite datasets consisted of 15 randomly sampled individuals from their respective locality. Each run was conducted with three replicates of four initial chains of 20,000 steps (1000 trees sampled every 20 reps, burn-in = 2000) and one final chain of 600,000 steps (30,000 trees sampled every 20 reps, burn-in = 30,000) using an adaptive heating scheme with four settings (1.0, 1.1, 1.2, 1.3). Priors for Θ ranged from 1.0×10^{-5} to 10.0 and priors for M ranged from 1.0×10^{-2} to 1000. Number of effective immigrants per generation for each population pair was estimated by multiplying M by its estimate of Θ for the recipient population (Kuhner 2006; Kuhner and Smith 2007). Differences in among tributary and within tributary migration rates between the two species were evaluated using Mann–Whitney U-tests.

DEMOGRAPHIC HISTORY

The summary statistics F_s (Fu 1997), R_2 (Ramos-Onsins and Rozas 2002), and Tajima's D (Tajima 1989) were calculated with mtDNA to test for departures from constant population size in the five clades of our phylogeographic analysis (Fig. 3). Of these statistics, F_s and R_2 have the greatest power to detect population growth, where growth is indicated by significantly large negative and small positive values of each, respectively (Ramos-Onsins and Rozas 2002). Significantly negative values of Tajima's D suggest growth or selective sweep, whereas the reverse may indicate selection, population subdivision, or recent population bottleneck (Tajima 1989). Estimates and corresponding significance of F_s, R_2, and Tajima's D were determined with 10,000 coalescent simulations in DnaSP.

To evaluate whether populations have experienced recent change in effective population size (N_e) based on microsatellites, we used M_P_Val.exe and Critical_M.exe (Garza and Williamson 2001) to compare the ratio of the number of alleles to the range in allele size (M) to population-specific critical M values (M_c). Briefly, M is expected to decrease with reduced N_e. For example, declining populations are expected to have M-ratios $< \sim 0.7$ due to the rapid loss of alleles when compared to the range in allele size, whereas more stable population sizes should have M-ratios closer to one (Garza and Williamson 2001). The M-ratio was estimated using a two-phase model with 90% single-step mutations, average size of non one-step mutations = 3.5, and prebottleneck Θ values of 10 ($N_e = 5000$). Loci that were monomorphic at the population level were removed from M-ratio analyses to avoid erroneous inflation of average M-ratios (Garza and Williamson 2001). The probability that a smaller M-ratio would be expected at equilibrium was assessed with 10,000 simulations.

Results

PHYLOGEOGRAPHICAL STRUCTURE AND DIVERGENCE TIME ESTIMATES

Of the 106 darters sequenced, 42 unique mtDNA haplotypes were identified (*E. boschungi* = 17, *E. tuscumbia* = 25; Table 1). The Bayesian analysis recovered *E. boschungi* and *E. tuscumbia* as sister with substantial mean mtDNA divergence between species (11.6%; Fig. 3). Four highly divergent clades were detected within *E. boschungi* (6.1% mean among clade sequence divergence), all but one of which was geographically structured according to distinct tributaries to the Tennessee River (Figs. 2, 3). Clade 2 (Figs. 2, 3) was the only exception, containing individuals from Shoal Creek and Buffalo River localities. The analysis recovered shallow divergence among all *E. tuscumbia* haplotypes (0.43% mean sequence divergence), with no apparent geographic structuring based on locality or tributary (Fig. 3). The statistical parsimony analysis recovered four disconnected clades for *E. boschungi* (not shown) that corresponded to clades defined in the MrBayes analysis (Figs. 2, 3). TCS connected all *E. tuscumbia* haplotypes with 95% confidence, and most low frequency or singleton haplotypes derived from the ancestral haplotype 26 or the highly abundant haplotype 25 (Fig. 3). Both haplotypes 25 and 26 are widespread and abundant in the central portion of the range of *E. tuscumbia* (Pryor Branch [PY], Limestone Creek [LM, TH, PK, BD], and Indian Creek [KL, BY, BR]; Figs. 1, 3; Table 1). However, each of these central localities had one or two uniquely derived haplotypes (Figs. 1, 3; Table 1). All localities at the margin of the range of

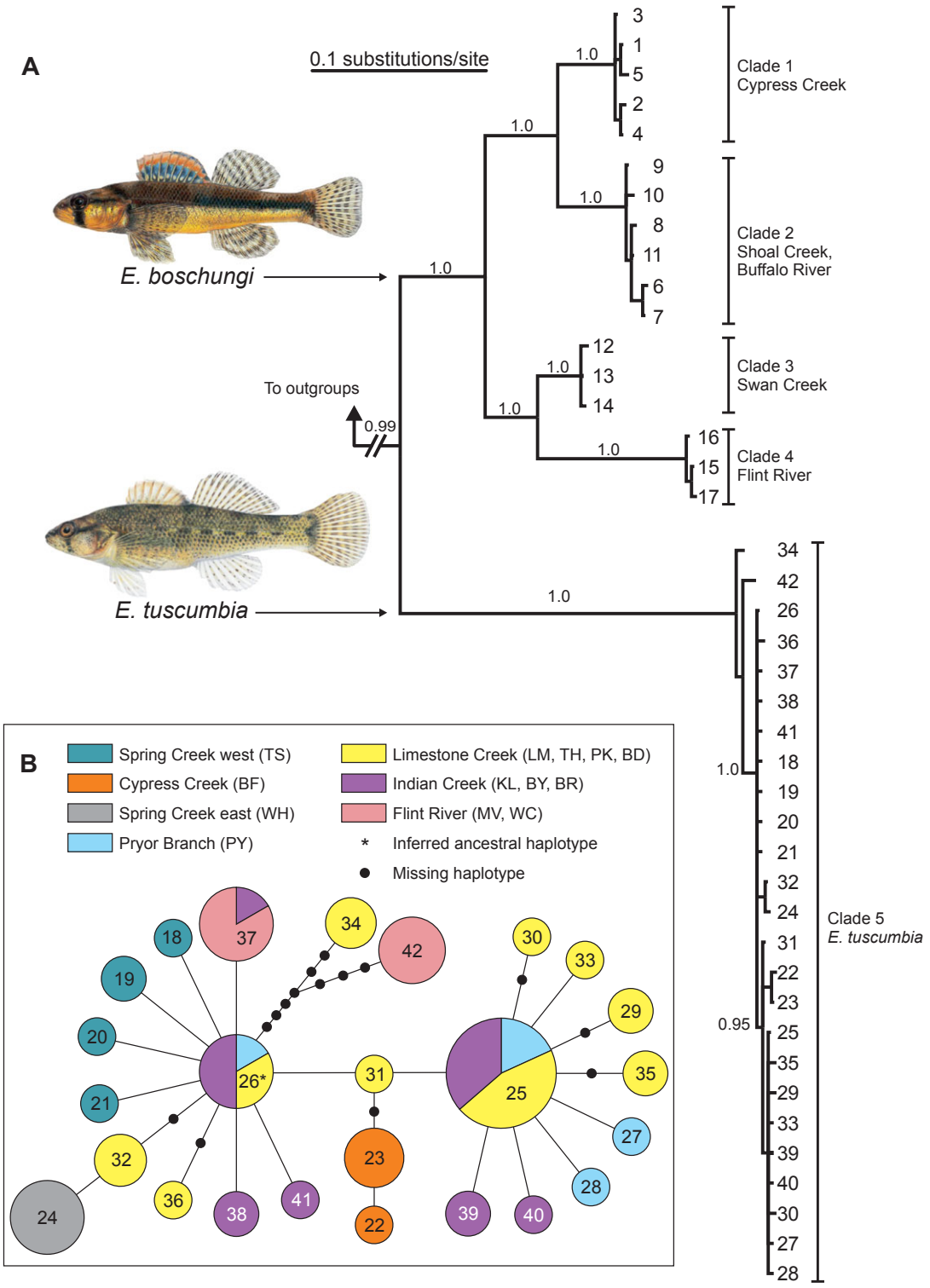

Figure 3. (A) Fifty percentage majority-rule consensus phylogram for Bayesian analyses of the mitochondrial (mt) ND2 dataset. Mean posterior probabilities shown at nodes with significant support (≥0.95) and haplotype codes at terminals follow Table 1. Outgroup taxa not shown for simplicity. (B) mtDNA haplotype network from the statistical parsimony analysis of *Etheostoma tuscumbia*. Circles (haplotypes) are proportional to sample size and lines indicate one mutational step between haplotypes. Haplotype and locality codes correspond to Table 1 and Figure 1.

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E. tuscumbia, that is, Cypress Creek, Spring Creek (west and east), and Flint River were characterized by unique, derived haplotypes (Figs. 1, 3; Table 1). Haplotypes 34 (PK) and 42 (MV) exhibited the highest degree of divergence from all other haplotypes, with six and seven mutational differences from the ancestral haplotype 26, respectively (Fig. 3).

Divergence time estimates yielded TMRCA for Centrarchidae (33.9 mya, 95% highest posterior density [HPD]: [27.2, 41.9]) consistent with previous estimates using this calibration method (Hollingsworth and Near 2009; Near et al. 2011). Divergence between *E. boschungi* and *E. tuscumbia* occurred during the late Miocene (8.1 mya [95% HPD: 5.6–10.8 mya]; Fig. 4). The TMRCA for the four clades of *E. boschungi* was estimated in the Pliocene (4.6 mya [95% HPD: 3.1–6.3 mya]; Fig. 4) and isolation of *E. boschungi* in distinct tributaries to the Tennessee River was estimated in the early Pleistocene (2.0 mya [95% HPD: 1.1–3.1 mya]; Fig. 4) for the clade containing Shoal Creek (SH) and Cypress Creek (DD), and in the late Pliocene (2.7 mya [95% HPD: 1.6–4.1 mya]; Fig. 4) for the clade containing Flint River (FL) and Swan Creek (SW). BEAST coalescent analysis indicated a root age of all *E. tuscumbia* in the late Pleistocene (0.5 mya [95% HPD: 0.2–0.7 mya]).

POPULATION GENETIC DIVERSITY, STRUCTURE, AND GENE FLOW

Of the nine microsatellite loci amplified for *E. boschungi*, EosC3 was monomorphic and EosD11 failed to amplify for all populations, thus both were removed from the dataset. MICRO-CHECKER indicated the possibility of null alleles for some populations at locus Esc120, but subsequent analyses with and without this locus were highly consistent. Thus, Esc120 was used in all final analyses of microsatellite data. Following Bonferroni correction, only three of the 39 population-locus comparisons for *E. boschungi* deviated from HWE and resulted from a deficiency of heterozygotes. Seven of the nine loci were successfully amplified for *E. tuscumbia*. Loci Eca48 and Esc120 failed to amplify across all populations and were removed from the dataset. Seven of the 80 tests for HWE deviated significantly following Bonferroni correction, all of which resulted from deficiency of heterozygotes. However, as with *E. boschungi*, there was no clear pattern of deviation from HWE within a single population and analyses with and without potentially problematic loci were highly congruent. No loci showed evidence of linkage.

Both *E. boschungi* and *E. tuscumbia* were similar for number of microsatellite alleles per locus (2−15 and 2−19, respectively). Measures of allelic variation (A, AR, H_o, and H_e) varied widely among localities for both species, with slightly lower mean values for each in *E. tuscumbia* compared to *E. boschungi* (Table 2).

The STRUCTURE analysis for *E. boschungi* recovered a high degree of population structure, with clusters corresponding to individual breeding sites or geographically proximate breeding sites ($K = 6$; Fig. 2). Analysis of *E. tuscumbia* also recovered high levels of genetic structure across the study area ($K = 8$; Fig. 1). However, clusters were more structured at the eastern and western margins of the range, and there was a high degree of admixture among localities within the central portion of the species' range (Fig. 1). Plots of the "estimated ln probability of the data" versus K and ΔK were unambiguous for $K = 6$ and $K = 8$ for *E. boschungi* and *E. tuscumbia*, respectively.

The AMOVAs were largely consistent with results of the phylogeographic and STRUCTURE analyses, in which *E. boschungi* exhibited a higher degree of among locality differentiation compared to *E. tuscumbia* across their respective ranges (Table 3). In AMOVAs that grouped populations among tributaries, differentiation was high and significant for *E. boschungi*, however, among tributary differentiation for *E. tuscumbia* was not significantly different from zero (Table 3).

Coalescent estimations of effective migrants per generation among tributaries were low for both *E. boschungi* (= 0.24 [microsatellites], = 0.02 [mtDNA]) and *E. tuscumbia* (= 0.31 [microsatellites], = 0.11 [mtDNA]), with only one comparison >1.0 for *E. tuscumbia* (Tables 4, 5). Migration within tributaries was higher for both *E. boschungi* (= 1.78 [microsatellites], = 0.51 [mtDNA]) and *E. tuscumbia* (= 2.19 [microsatellites], = 0.95 [mtDNA]), with several comparisons among localities >1.0 for both species (Tables 4, 5). MtDNA-based migration rates among tributaries were significantly lower in *E. boschungi* when compared to *E. tuscumbia* ($P = 0.014$). Microsatellite-based migration rates among tributaries were not significantly different between species.

DEMOGRAPHIC HISTORY

Coalescent estimations of F_s, R_2, and Tajima's D for the four clades of *E. boschungi* (Fig. 3) were not significant, indicating no evidence of departure from constant population size. Only F_s (–9.39, $P < 0.001$) was significant for *E. tuscumbia* (clade 5; Fig. 3), indicating a signature of population growth. Microsatellite-based M-ratios of all *E. boschungi* populations were well above critical values, indicating no evidence for recent reductions in N_e (Table 2). In contrast, M-ratios of three *E. tuscumbia* populations on the eastern and western margins of the species range were significantly below critical M values, indicating recent or prolonged reductions in N_e (Table 2). In addition, M-ratios of one marginal and two centrally located populations of *E. tuscumbia* were low enough to support genetic bottleneck ($M < 0.68$; Garza and Williamson 2001) despite being nonsignificant with respect to their respective critical values (Table 2).

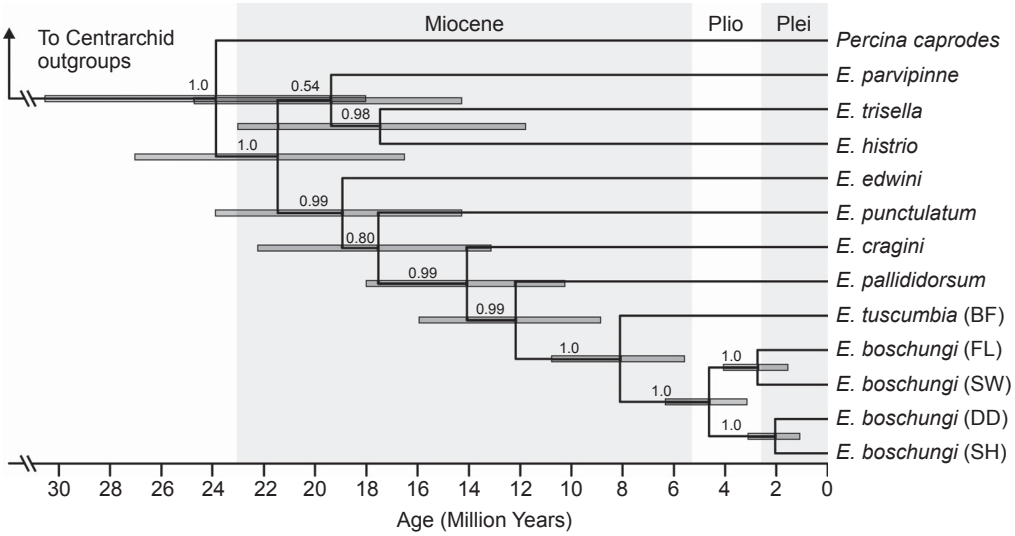

Figure 4. Chronogram for darter species (pruned from the BEAST analysis with centrarchid outgroups) based on the combined analysis of mitochondrial ND2 and nuclear S7 gene sequences. Posterior probabilities are shown at nodes. Gray bars at nodes represent the 95% HPD of age estimates. Taxon names are followed by codes corresponding to Table 1 and Figures 1 and 2.

Discussion

Establishing a link between the ecology and evolution of species is crucial to understand factors important in lineage diversification (Bohonak 1999). Previous studies have articulated the importance of habitat preference, as it relates to life history, in differentiation of aquatic taxa (Tringali et al. 1999; Rocha et al. 2002; Marten et al. 2006; Papadopoulou et al. 2009) and to some extent freshwater fishes (Tibbets and Dowling 1996; Turner et al. 1996; Turner and Trexler 1998; Whiteley et al. 2004). Our analysis of sister, co-distributed darters with differing breeding strategies revealed highly incongruent phylogeographic histories, patterns of population structure and migration, and demographic histories between the two species. These findings indicate that life-history strategies involving strict breeding habitat requirements in headwaters may impose a strong influence on dispersal ability and gene flow in darters, mainly by limiting dispersal in large river courses. We discuss below how extrinsic factors, namely spatiotemporal habitat stability, have likely interacted with intrinsic factors in generating and maintaining the patterns of genetic structure within both species examined.

BARRIERS TO DISPERSAL AND GENETIC DIFFERENTIATION

Large streams and rivers are thought to pose strong barriers to dispersal for spring inhabiting fishes of the southeastern United States (Starnes and Etnier 1986). In addition, recent studies suggest that darters with highly specific reproductive strategies, particularly in headwaters, can experience microallopatric divergence or reduced gene flow among populations separated by large river courses (Turner and Robison 2006; Hollingsworth and Near 2009). Given the shared geographic distribution of *E. boschungi* and *E. tuscumbia* in tributaries to the Tennessee River, their sister relationship, and strict reliance on groundwater habitats, we predicted congruent phylogeographic structure for both species and that both species would have greater population differentiation among tributaries to the Tennessee River versus within. Given the strict breeding habitat requirements of *E. boschungi*, we further predicted greater differentiation both among and within tributaries compared to *E. tuscumbia*. The AMOVA, STRUCTURE, and migration analyses indicated that *E. boschungi* exhibited higher levels of among-population differentiation compared to *E. tuscumbia* (Table 3). Both species exhibited local signs of genetic structure within the same tributary, which is common for spring endemic fishes of this region (Fluker et al. 2010; Fluker 2011). However, when compared among tributaries to the Tennessee River, significant differentiation was only detected for *E. boschungi* (Table 3), indicating that the major river course is a barrier for *E. boschungi*, but not for *E. tuscumbia*.

Our phylogeographic and divergence dating analyses further supported that the Tennessee River poses a barrier to gene flow for the more specialized *E. boschungi*, but not for *E. tuscumbia*. *Etheostoma boschungi* is clearly structured among tributaries to the Tennessee River with a lack of gene flow among these tributaries since the late Pliocene or early Pleistocene (Figs. 2, 4). In contrast, all *E. tuscumbia* haplotypes share a common ancestor in the late Pleistocene and we found no clear pattern of phylogeographic structuring among localities or tributaries to the Tennessee River (Figs. 1, 4). These differences refute our prediction and were unexpected based on patterns seen in other

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Table 2. Genetic diversity estimates (averaged over seven microsatellite loci) for *Etheostoma boschungi* and *E. tuscumbia*.

Species/population	mtDNA Clade	N	A	AR	H_o	H_e	M_c	M-ratio	P-value M-ratio
E. boschungi									
CB	1	31	3.43	3.34	0.48	0.46	0.66	0.79	0.598
DD	1	27	4.71	4.66	0.54	0.55	0.65	0.74	0.320
SH, CH	2	10	4.00	4.00	0.52	0.59	0.56	0.77	0.790
NF, SF	2	29	6.57	6.13	0.67	0.64	0.65	0.71	0.208
SW	3	21	8.43	8.43	0.73	0.74	0.62	0.75	0.465
FL	4	31	10.00	9.28	0.80	0.85	0.66	0.86	0.906
Population mean	–	–	6.74	6.50	0.65	0.67	–	–	–
E. tuscumbia									
TS	5	24	7.29	6.51	0.71	0.64	0.64	0.63**	0.047
BF	5	22	1.86	1.74	0.25	0.16	0.60	0.63*	0.095
WH	5	24	4.86	4.50	0.60	0.61	0.64	0.75	0.436
PY	5	24	7.86	7.08	0.64	0.68	0.64	0.73	0.325
LM	5	20	5.43	5.25	0.61	0.65	0.63	0.66*	0.101
TH	5	22	6.00	5.60	0.62	0.62	0.64	0.69	0.174
PK	5	24	6.14	5.67	0.58	0.60	0.65	0.67*	0.090
BD	5	23	8.29	7.50	0.65	0.68	0.64	0.79	0.677
KL	5	24	4.57	4.42	0.45	0.56	0.64	0.73	0.353
BY	5	16	7.29	7.29	0.65	0.69	0.62	0.77	0.672
BR	5	24	5.29	4.96	0.67	0.63	0.63	0.76	0.483
MV	5	24	2.29	2.22	0.40	0.28	0.61	0.43**	0.000
WC	5	24	4.67	4.03	0.57	0.56	0.64	0.62**	0.029
Population mean	–	–	5.53	5.14	0.57	0.57	–	–	–

N = number of individuals; A = mean number of alleles per locus; AR = allelic richness; H_o = heterozygosity observed; H_e = heterozygosity expected. Genetic diversity estimates are followed by critical M values (M_c), M-ratios, and associated significance as performed using the methods of Garza and Williamson (2001). See Figure 3 for clade membership. Populations described and mapped in Table 1 and Figures 1 and 2.

*Value suggests bottleneck (<0.68), but not significant.

**Value significant.

spring inhabiting darters of the region. For example, the Watercress Darter (*E. nuchale*) and the Rush Darter (*E. phytophilum*) both exhibit high degrees of phylogeographic structure among spring populations distributed in small tributaries to the Black Warrior River of the Mobile Basin (Fluker et al. 2010; Fluker 2011).

The odd occurrence of *E. boschungi* in tributaries to the Buffalo River provides a unique insight into the dispersal mechanisms for this species. The distribution of suitable breeding habitats for *E. boschungi* is highly discontinuous and determined by stream order and local geological conditions. For example, breeding habitats are ephemeral tributaries of small order streams within the Fort Payne Chert formation of the Interior Low Plateaus Physiographic Province (Boschung and Nieland 1986; Osborne et al. 1998.; Szabo et al. 1988; McGregor and Shepard 1995). The propensity of *E. boschungi* to migrate into these seeps is great and breeding adults have been found at the most upstream reaches where water depths may be as little as 4 cm (Boschung and Nieland 1986). These ephemeral breeding tributaries closely interdigitate among the low divides separating the Buffalo River and Shoal Creek and it is likely that rare chance events such as stream capture or flooding allow brief periods of dispersal between drainage basins. Microsatellite and mtDNA data support very recent or ongoing gene flow between proximate localities in Shoal Creek (SH) and the Buffalo River (CH) and all localities within the Buffalo River (CH, NF, SF) and Shoal Creek (SH) were recovered in clade 2 (Fig. 2). Thus, intertributary dispersal in *E. boschungi* is likely determined by proximity of favorable breeding sites that interdigitate among drainage divides, rather than dispersal through larger order streams and rivers.

Dispersal mechanisms are much different for *E. tuscumbia*. That the Tennessee River was not a significant barrier to dispersal was surprising for this spring specialist. That all extant populations of *E. tuscumbia* recently stemmed from the more diverse, centrally located populations supports the extraordinary dispersal ability of this spring endemic species. With the exception of locality BD, which falls within the Fort Payne Chert Formation, the location of suitable spring habitats for *E. tuscumbia* lies within

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Table 3. Analysis of molecular variance (AMOVA) for different hierarchical groupings of *Etheostoma boschungi* and *E. tuscumbia* based on the mitochondrial (mt) ND2 gene and seven microsatellite loci. Nonsignificant results (>0.05) are shown in bold.

Species/data type	Source of variation	d.f.	SS	Percentage of variance	Fixation index			*P*-value
E. boschungi								
mtDNA	Among localities	7	831.47	98.04	Φ_{ST}	=	0.98	<0.00001
	Within localities	33	15.27	1.96	–			
Microsatellites	Among localities	7	243.10	29.20	F_{ST}	=	0.29	<0.00001
	Within localities	141	327.06	3.79	F_{IS}	=	0.05	<0.01
	Within individuals	149	310.50	67.01	F_{IT}	=	0.33	<0.00001
mtDNA	Among tributaries	3	807.87	94.91	Φ_{CT}	=	0.95	<0.00001
	Among localities within tributaries	4	23.60	3.57	Φ_{SC}	=	0.70	<0.00001
	within localities	33	15.27	1.52	Φ_{ST}	=	0.98	<0.01
Microsatellites	Among tributaries	3	162.50	9.66	F_{CT}	=	0.10	<0.05
	Among localities within tributaries	4	80.61	20.63	F_{SC}	=	0.23	<0.00001
	within localities	290	637.56	69.70	F_{ST}	=	0.30	<0.00001
E. tuscumbia								
mtDNA	Among localities	12	99.88	63.17	Φ_{ST}	=	0.63	<0.00001
	Within localities	52	45.20	36.83	–			
Microsatellites	Among localities	12	382.65	25.21	F_{ST}	=	0.25	<0.00001
	Within localities	282	583.32	4.53	F_{IS}	=	0.06	<0.00001
	Within individuals	295	540.50	70.26	F_{IT}	=	0.30	<0.00001
mtDNA	Among tributaries	6	56.53	10.53	Φ_{CT}	=	0.11	**0.32327**
	Among localities within tributaries	6	43.35	53.14	Φ_{SC}	=	0.59	<0.00001
	within localities	52	45.20	36.33	Φ_{ST}	=	0.64	<0.00001
Microsatellites	Among tributaries	6	238.94	7.05	F_{CT}	=	0.07	**0.13366**
	Among localities within tributaries	6	143.62	18.91	F_{SC}	=	0.20	<0.00001
	within localities	577	1123.82	74.05	F_{ST}	=	0.26	<0.00001

d.f. = degrees of freedom; SS = sum of squares.

the Tuscumbia Limestone formation, which adheres closely to the valley floor of the Tennessee River (Osborne et al. 1998.; Szabo et al. 1988; Etnier and Starnes 1993). Numerous known, and probably many unknown, spring localities for *E. tuscumbia* were inundated with the construction of Pickwick Reservoir in 1938 (Fig. 1), indicating a formerly more widespread distribution of this species and its habitats (Etnier and Starnes 1993; Mettee et al. 1996; Boschung and Mayden 2004). Thus, historic spring to spring dispersal may have involved brief forays along the margins of the Tennessee River when springs were much more abundant along the valley floor.

INTRINSIC CONTRIBUTIONS TO GENETIC DIFFERENTIATION AND LINEAGE DIVERGENCE

Turner and Trexler (1998) determined that body size, egg size, and fecundity exert a significant influence on gene flow in darters. Unfortunately, few studies have evaluated these characteristics for *E. boschungi* and *E. tuscumbia*. Although these life-history characters were not the focus of our study, there is some evidence that maximum body size, egg size, and fecundity are similar or overlapping between the two species (Table S3).

With the exception of *E. boschungi* and *E. tuscumbia*, no other sympatric sister pair of darters exhibit such vast differences in reproductive strategy. The unique breeding migration behavior by *E. boschungi* into patchily distributed ephemeral habitats, and the short timeframe in which reproductive migrations occur, are presumably key intrinsic factors that limit dispersal ability and gene flow, and thus contributed to fine-scale diversification within this species. Adult *E. boschungi* complete their life history in two distinct habitats within a small geographical area (Boschung and Nieland 1986), which places strict requirements on their breeding success. Because the spawning habitat is nonexistent (dry) throughout most of the year, reproductively mature adults must remain in streams within their home range during nonspawning times if they are to successfully reproduce. Adults that migrate outside of their home range either fail to successfully navigate large river courses between breeding habitats in neighboring tributaries, fail to locate breeding habitats within their tributary, or arrive at spawning habitats outside of the one- to two-month spawning window. Although homing to natal spawning grounds has not been examined for *E. boschungi*, microsatellite DNA analyses of *E. trisella*, which exhibits the same spawning strategy, suggest that adults return to natal sites to spawn at a higher

Table 4. Rates of migration (*M*) for *Etheostoma boschungi* as estimated in LAMARC.

	Microsatellites		mtDNA	
Comparison	*M*	4*N*m	*M*	*N*m
Among tributaries				
CB -> SH/CH	0.088	0.186	29.320	0.014
CB -> SW	0.038	0.324	0.043	0.000
CB -> FL	0.019	0.140	1.415	0.002
SH/CH -> CB	0.119	0.170	26.910	0.053
SH/CH -> SW	0.044	0.374	1.090	0.003
SH/CH -> FL	0.017	0.131	7.890	0.009
SW -> CB	0.080	0.114	5.513	0.011
SW -> SH/CH	0.085	0.179	9.897	0.005
SW -> FL	0.057	0.434	26.111	0.030
FL -> CB	0.054	0.077	2.330	0.005
FL -> SH/CH	0.042	0.089	7.738	0.004
FL -> SW	0.071	0.611	21.827	0.062
Mean	0.060	0.236	11.674	0.017
Within tributaries				
NF -> SF	0.372	1.478	568.899	0.307
SF -> NF	1.421	4.973	780.948	0.772
CB -> DD	0.157	0.312	692.741	0.515
DD -> CB	0.272	0.359	419.806	0.461
Mean	0.556	1.781	615.599	0.514

Number of effective migrants per generation (4*Nm*) is shown for microsatellites and number of effective females migrating per generation (*N*m) is shown for mitochondrial (mt) DNA. Locality codes are described and mapped in Table 1 and Figure 2.

Table 5. Rates of migration (*M*) for *Etheostoma tuscumbia* as estimated in LAMARC.

	Microsatellites		mtDNA	
Comparison	*M*	4*N*m	*M*	*N*m
Among tributaries				
BF -> WH	0.048	0.096	146.917	0.012
BF -> BD	0.034	0.243	85.785	0.376
BF -> MV	0.115	0.073	15.221	0.005
WH -> BF	0.141	0.061	15.539	0.005
WH -> BD	0.181	1.293	98.860	0.434
WH -> MV	0.266	0.169	15.984	0.005
BD -> BF	0.190	0.083	462.550	0.159
BD -> WH	0.246	0.491	793.380	0.063
BD -> MV	0.423	0.269	137.900	0.044
MV -> BF	0.070	0.030	140.558	0.048
MV -> WH	0.124	0.247	48.830	0.004
MV -> BD	0.095	0.676	46.343	0.203
Mean	0.161	0.311	167.322	0.113
Within tributaries				
LM -> BD	0.282	1.647	910.780	2.806
BD -> LM	0.359	1.206	906.924	3.848
MV -> WC	0.061	0.139	102.920	0.026
WC -> MV	0.128	0.115	127.409	0.024
KL -> BR	0.174	0.475	279.673	0.217
KL -> BY	0.685	3.889	246.730	0.373
BR -> KL	0.189	0.734	337.010	0.299
BR -> BY	1.630	9.261	627.070	0.947
BY -> KL	0.854	3.311	435.514	0.386
BY -> BR	0.398	1.086	729.434	0.566
Mean	0.476	2.186	470.346	0.949

Number of effective migrants per generation (4*Nm*) is shown for microsatellites and number of effective females migrating per generation (*N*m) is shown for mitochondrial (mt) DNA. Locality codes are described and mapped in Table 1 and Figure 1.

frequency than nonnatal sites within a watershed (M. S. Piteo et al., unpubl. data).

The extraordinary dispersal ability of *E. tuscumbia* was unknown previous to this study. We suggest that extrinsic factors such as long-term habitat availability and stochastic processes throughout the Pleistocene (see below) represent avenues for further research into the evolution of dispersal ability for *E. tuscumbia*. Although *E. tuscumbia* exhibits a strict habitat preference in spring pools, the breeding strategy is less specific (i.e., no migratory behavior into ephemeral habitats and relaxed seasonal timing) compared to *E. boschungi*. The protracted spawning period for *E. tuscumbia* (Table S3) is likely a key life-history trait that interacts with dispersal to enhance gene flow within this species. For example, individuals that successfully migrate to other suitable spring habitats may be able to settle and find receptive mates regardless of season. In the absence of a strict spawning window, immigrants may have an immediate impact on gene flow, in addition to reduced mortality and deferred costs of dispersal (Baguette and Van Dyck 2007).

In this study and other comparisons of multiple co-distributed freshwater fish species with differing intrinsic characteristics, the emerging trend is that population divergence and lineage diversification is greater in those species with intrinsic characteristics conducive to fragmentation (Tibbets and Dowling 1996; Turner and Trexler 1998; Turner and Robison 2006). This trend is also evident within the *E. basilare* complex, which is restricted to headwater streams with suitable "slab-rock" breeding habitat and exhibits deep lineage divergence among tributaries within a single river (Page 1983; Hollingsworth and Near 2009). In these cases, strict breeding strategies entrench populations into small fragmented patches with suitable breeding habitat and individuals are poorly suited to successfully navigate interconnecting habitats. For darters, it appears that specialized breeding behaviors in headwater habitats may be an important intrinsic characteristic that promotes diversification on fine scales where there are no apparent physical barriers to gene flow.

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EXTRINSIC CONTRIBUTIONS TO GENETIC DIFFERENTIATION AND LINEAGE DIVERGENCE

The strikingly different patterns of genetic structure between *E. boschungi* and *E. tuscumbia* indicate that life-history strategy influences dispersal ability, genetic differentiation, and lineage diversification within these species. However, their differing demographic histories point to possible differences in habitat stability, and ignoring extrinsic factors such as drainage history may confound the ability to understand the current patterns of genetic structure (Tibbets and Dowling 1996; Burridge et al. 2008). Thus, differences in genetic structure and demographic history between these two species must also be understood in the context of the complex paleohydrologic history of the lower bend of the Tennessee River.

The star-like patterns observed in the mtDNA haplotype network and the significantly negative F_s values for *E. tuscumbia* are indicative of recent population growth (Fu 1997; Avise 2000; Fig. 3). However, several marginal populations of *E. tuscumbia* have passed through recent or prolonged bottlenecks based on microsatellite data (Table 2). Additionally, marginal populations exhibit reduced mtDNA haplotype diversity, higher genetic differentiation, and generally lower levels of genetic variation based on microsatellites (Figs. 1, 3; Tables 1, 2). Collectively, these patterns are indicative of a history of frequent extinction and colonization events (Pannell 2003), indicating that springs inhabited by *E. tuscumbia* were unstable throughout the Pleistocene.

Instability of springs inhabited by *E. tuscumbia* could have occurred during the Pleistocene in two main ways. Although the lower bend of the Tennessee River was not directly impacted by Pleistocene glacial ice sheets, climate fluctuations during this time had a dramatic effect on species range dynamics in North America (Hewitt 2011). The TMRCA of *E. tuscumbia* roughly corresponds to the transition from relatively shorter Pleistocene glacial cycles (≈41,000 years long) to relatively longer, more dramatic cycles (≈100,000 years long) approximately 0.9 mya (Hewitt 2000, 2011). During climate fluctuations, dryer periods could have resulted in local extinction of springs or range contraction, whereas wetter periods could have caused local reconfigurations of spring out flows, facilitating dispersal and range expansion (Williams 1968).

An alternative explanation can be drawn from hypothesized course reconfigurations of lower portions of the Tennessee River during the Pleistocene (Thornbury 1965; Braasch and Mayden 1985; Starnes and Etnier 1986). Although these reconfigurations were hypothesized to occur downstream of our study area, they could have drastically altered the flow regime of the Tennessee River by inducing periods of incision and aggradation. Consequently, aquifers closely associated with the river system would have been heavily impacted by such events (Creuzé des Châtelliers et al. 1994). Drill cores from the lower Tennessee River identified Pleistocene valleys buried at least 21 m below the present valley depth, and groundwater levels were equally depressed during this time (Moneymaker 1941). Thus, groundwater habitats closely aligned with the Tennessee valley (Tuscumbia Limestone formation) may have been more heavily impacted during the Pleistocene, and those at higher elevations and further distance from the valley (Fort Payne Chert formation) may have served as refugia during times of instability. Our data support the existence of two or three refugia for *E. tuscumbia* during the late Pleistocene. For example, the most upstream localities in Indian Creek (KL) and Flint River (WC) were resolved as a homogeneous cluster in the STRUCTURE analysis and WC is monomorphic for an mtDNA allele present in KL (Fig. 1), suggesting a recent connection between these two disjunct upstream localities or relicts of a recent shared distribution. The two most divergent haplotypes recovered in the analysis (34 and 42; Fig. 3; Table 1) were found at the central locality PK and an upstream locality MV, possibly reflecting relict haplotypes retained in refugia. Interestingly, Beaverdam Spring (BD) in the Limestone Creek drainage is one of the largest spring wetlands in the region and the only spring inhabited by *E. tuscumbia* that falls outside of the Tuscumbia Limestone formation. Accordingly, this locality exhibits the highest levels of genetic variation and houses the inferred ancestral mtDNA haplotype (Fig. 3; Table 2). The co-occurrence of two other spring endemic fishes provides further evidence for long-term stability of spring habitats within the range center of *E. tuscumbia*. The Spring Pygmy Sunfish (*Elassoma alabamae*) was once known from three tributaries to the Tennessee River, but now is only known from the Beaverdam Creek system (Mayden 1993; Localities BD and TH, Fig. 1). The Whiteline Topminnow (*Fundulus albolineatus*) was known only from Big Spring (Indian Creek; Fig. 1), but is now extinct due to recent anthropogenic disturbance (Williams and Etnier 1982).

Although we cannot confidently distinguish which mechanism(s) affected range dynamics of *E. tuscumbia* during the Pleistocene, our results indicate periods of extinction and recolonization throughout the range of this species. There is further evidence that areas within its range center were more stable during the Pleistocene and may have served as refugia during instability.

We found no evidence for population expansion or prolonged bottlenecks within *E. boschungi*, suggesting relatively constant long-term population sizes throughout the Pleistocene, or that populations have recovered from any historic reductions in N_e. Constant long-term N_e, pre-Pleistocene divergence among tributaries, and fine-scale genetic structure among *E. boschungi* populations indicate that suitable breeding and nonbreeding habitats were relatively stable throughout the Pleistocene. Breeding sites occupied by *E. boschungi* are situated in an area of long-term

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geologic stability (Thornbury 1965; Mayden 1988), and are located a considerable distance from the valley floor of the Tennessee River. Thus, aquifers associated with *E. boschungi* breeding sites would have experienced little or no impacts from reconfigurations of the lower Tennessee River.

Conclusions

Darters represent a model group to better understand the pattern-process relationships that drive freshwater fish diversification on fine geographic scales. This study highlights the use of a rare example of sister, co-distributed darters with differing breeding strategies to examine the effects of life-history strategy on phylogeographic structure, gene flow, and lineage diversification in a setting with no obvious extrinsic barriers to gene flow. Our findings indicate that life-history strategy may impose a strong influence on dispersal ability and gene flow in darters with strict breeding habitat requirements in headwaters, mainly by limiting dispersal in large river courses. The idea that microallopatric divergence, driven by specialized habitat preference and isolation in tributaries within a river system, is a driver of speciation in darters (see Hollingsworth and Near 2009) is inherent from the patterns seen in *E. boschungi*. The four highly divergent clades within *E. boschungi* represent previously unrecognized diversity and potentially three to four cryptic species. Although spatio-temporal habitat stability has likely contributed to patterns of genetic differentiation in our study species, our results suggest that life-history strategy may play a more important role in population subdivision and lineage diversification in darters than previously appreciated.

ACKNOWLEDGMENTS

We thank A. R. Henderson, J. H. Howell, C. E. Johnston, M. Sandel, B. W. Stallsmith, and M. P. Venarsky for assistance with tissue collections. P. L. Rakes, J. R. Shute, and R. L. Mayden graciously provided additional tissue samples. R. Cassidy and S. Cassidy provided laboratory assistance. Fish illustrations were used with permission from J. R. Tomelleri. We thank the Alabama Department of Conservation and Natural Resources (ADCNR), National Park Service, and Tennessee Wildlife Resources Agency for fish collecting permits. Nonlethal tissue samples of *E. boschungi* were obtained under permit number TE171564-0 (U.S. Department of Interior). This project was supported by a State Wildlife Grant (ADCNR).

DATA ARCHIVING

DNA sequence data available from GenBank (accession numbers JN972486-JN972533) Microsatellite DNA data available from the Dryad digital repository: http://doi.org/10.5061/dryad.2s07g.

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Associate Editor: T. Near

Supporting Information

Additional Supporting Information may be found in the online version of this article at the publisher's website:

Table S1. GenBank accession numbers for the mitochondrial ND2 gene and the nuclear ribosomal S7 intron 1 for all taxa used in this study, followed by University of Alabama Ichthyological Collection (UAIC) catalog numbers for tissue and voucher specimens.
Table S2. Best-fit models of molecular evolution for character partitions used in MrBayes and BEAST analyses.
Table S3. Life-history characteristics for *Etheostoma boschungi* and *E. tuscumbia*.

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